

Scale AI With Token Unit Economics (Tokenomics), Not Budget Forecasts

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Initiatives: Functional Design; Redesign IT Operating Model For Effectiveness In Turbulent Times

CIOs who struggle to align the business demand with the cost of supply risk AI cost blowouts. CIOs must rethink predictive budgeting and vendor contracts by shifting to models that link variable AI token use to measurable business outcomes.

Insights at a Glance

AI introduces highly variable, probabilistic usage-driven costs that cannot be forecast accurately. Traditional budgeting and contracting practices attempt to force predictability through estimates and token commitments, often distorting decisions and constraining value.

Key Advice

CIOs must move away from predictive budgeting toward a more proactive and strategic operating model that aligns AI spending to business outcomes. AI consumption should be governed as a variable, value-linked investment, not a fixed, forecast-driven cost.

Core Insights

- The highest-value AI use cases are inherently unpredictable; constraining usage limits outcomes.
- Forecast-driven funding misallocates spend, constraining high-value use while tolerating low-value activity.
- Token consumption reflects the cost of supply, not the business value; outcome-based evaluation is essential.
- Treating AI as an IT cost obscures accountability and weakens business decision making.

Recommended Actions

- Replace upfront forecasts with continuous tracking of cost per outcome.
- Attribute AI costs to business use cases to enable value-based decisions.
- Prioritize high outcome-to-cost ratios to justify variable spend.

Adopt agile, business-led governance to scale AI usage in line with realized value.

Impact

AI introduces highly variable, usage-driven costs that cannot be accurately forecast in advance. In response, many organizations attempt to force predictability through token bundles, contracts, and estimation tools — creating a false sense of control until the production invoices arrive. Poor decisions result, where high-value AI use cases are constrained and low-value usage is tolerated simply because it fits budget expectations. CIOs then risk underinvesting in high-value AI opportunities, overspending on low-value usage, and losing credibility with business and finance stakeholders when forecasts inevitably fail.

Actions

CIOs must ensure that their teams:

- Replace upfront cost forecasting with continuous tracking of cost per outcome, enabling teams to scale AI usage based on realized business value rather than predicted spend.
- Empower business users to make informed decisions about how much they spend on each AI use case by adapting the technology operating model to make token consumption a business-controlled investment, not an unattributed spend item lost in an IT cost center.
- Prioritize high outcome-to-cost ratios to justify budget variance so that, regardless of how many tokens are consumed, the outcome is valuable enough to justify proportionate AI spending.

Cautions

- AI contracts often foster the illusion of predictability by including fixed-cost commitments that don't scale with token consumption or business outcomes and skew cost-value ratios.
- Nonprofit, government or regulated industries on fixed incomes cannot scale and fund costs in proportion to outcomes unless they prioritize their mission over financial responsibility.

How to Execute

Adopt a Consumption-Based Operating Model

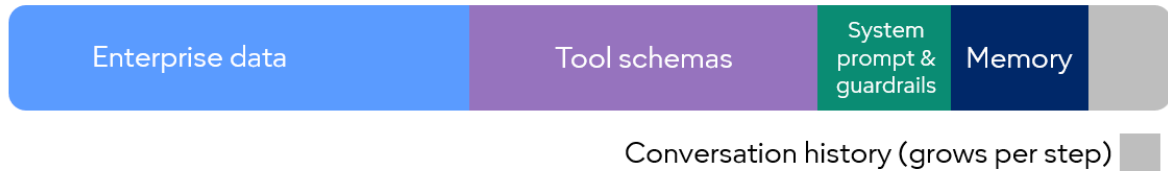
AI cost is variable and needs to be managed in proportion to value. AI creates value by adapting to changing conditions by making its cost inherently variable. CIOs need an operating model that doesn't rely on forecast accuracy but enables stakeholders to continuously manage the cost of AI in proportion to value. AI tokenomics compare the business value of an AI use case with the cost of tokens consumed (see [AI Tokenomics: Measuring LLM Usage in AI-Enabled Applications](#)).

The study of token economics, originally blockchain and cryptocurrency tokens, has been extended to AI. ¹ A **large language model** (LLM) breaks complex words down into components for AI to process, each represented by an input token. The consumption of other resources is also measured in tokens. Better reasoning capabilities consume more tokens, as confirmed by leaked documents. ² Vendors cannot afford to charge per user when users consume an ever-increasing number of tokens. Although vendors may define, consume and charge for tokens differently, tokens track their cost of supply, as illustrated in Figure 1.

Figure 1: Token Cost Composition per AI Agent Action

Token Cost Composition per AI Agent Action

INPUT TOKENS



OUTPUT TOKENS



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Trying to convert developer time into tokens is challenging.³ There's no simple conversion ratio, and token forecasts are statistically proven to be inaccurate. Tokens are also ineffective at measuring the value of the output. Creating value by developing and testing innovative software is a risky business that doesn't always pay off. AI can improve the odds of success, but at the risk of greater cost unpredictability. CIOs must act as a catalyst to drive higher value per token to compensate for overoptimistic estimates to ensure that risky bets pay off. Variable costs present the greatest risk to organizations with fixed or constrained income such as nonprofit, government and regulated organizations.

Use estimation tools such as the [Microsoft Copilot Credit Estimator](#) or [Amazon Bedrock AgentCore](#) to set expectations that future cost or value cannot be reliably predicted. This changes the CIO and CFO relationship from an annual planning meeting to a real-time partnership. Real value is only understood through usage. Treat pilots and early deployments as proofs of value (POVs) so that the business can adjust funding in response to observed outcomes in place of forecasts and assumptions. Start building an "operating system" to manage the new operating model.

Continuously monitor and adapt AI usage to deliver more value than cost, using a combination of agile, democratized and business-led ways of working to validate and adapt assumptions. Empower the relevant technology, business and finance experts to investigate spikes in cost that don't deliver value and take remedial action.

Make AI Cost a Business Cost

AI costs are driven by business usage but charged to IT. CIOs must ensure that AI costs are clearly tagged with identifiers, so they can be charged to the business use cases that generate them. Accounts payable functions typically ask IT to pass on the AI bill because they can't. Some AI technology provider invoices already follow the **FinOps Open Cost and Usage Specification (FOCUS)**,⁴ while others still need encouragement to follow. FOCUS enables organizations to tag usage with identifiers that can be used to pass on the costs of business usage through cloud financial management tools (see [Critical Capabilities for Cloud Financial Management Tools](#)). However, AI token consumption is typically far more detailed than cloud infrastructure consumption, as described in [Beyond Workflow Automation: When to Agentify With AI](#). Business-led AI adoption without coding in the necessary identifiers makes **value stream mapping** challenging. CIOs must mandate tags for cost transparency.

A \$1,000 AI use case justifies fewer tokens than a \$1 million one. However, risk-based businesses such as insurance companies may be willing to pay more to identify and prevent 1,000 fraudulent \$1,000 transactions.

This example illustrates how IT spend is assessed by the value of the outcome, not the number of tokens used. CIOs must enable business executives to make these important business decisions because AI that runs a business process is a business cost, not an IT cost. That's why business process outsourcing (BPO) is explicitly excluded from Gartner's IT budget framework definitions (see [IT Key Metrics Data 2026: Industry Measures — Framework Definitions](#)).

Design Cost Controls to Meet Strategic Objectives

Tokenmaxxing is where human workers consume the maximum number of tokens (see [Analyst Take: Making the Case for Tokenmaxxing](#)). This is because managers confuse higher consumption with higher productivity even when it doesn't improve outcomes. Fewer tokens may be used by a smaller, context-specific AI model, such as in cybersecurity (see [Emerging Tech: Small Language Model Precision, Not LLM Scale, Redefines Cybersecurity Cost and Efficacy](#)). While usage can be limited, for example, by tuning the context or limiting reasoning in LLMs, this risks limiting value.

Both internal and external providers have been reluctant to limit AI adoption for this reason. However, few CIOs or AI providers can afford to subsidize the business use of AI forever. Some CIOs or business executives seek to drive AI adoption by removing cost guardrails and subsidizing the cost. Few organizations can afford to subsidize the business adoption of AI for long, but some nonprofit organizations justify AI spending.

An insurer used AI to price risk and quote insurance premiums. This worked brilliantly until AI consumption spiked and policy sales stalled. Investigation revealed competitors were asking increasingly complex questions to beat its insurance quotes. But instead of rationing prospective customers to three questions, the insurer kept token consumption proportional to genuine sales opportunities by eliminating nonhuman requests and simplifying prompts. It continued to scale usage because the business value outweighed the additional cost — showing that value, not forecasts, should guide decisions.

Switch From Tokenmaxxing to Valuemaxxing

The priority is not to predict the cost but to maximize the value of AI so that it covers the cost. CIOs must focus teams on designing and testing specific use cases for AI to ensure that business outcomes are worth more than they cost. Take the example of an agent that analyzes system logs for issues and tunes the system to maximize performance. Nobody can predict how many log entries will require analysis. Whether AI processes 1,000 or 1 million entries is less important than the value of the insights it generates.

Pilot studies are often conducted for ideal use cases in sheltered test environments by expert teams as a priority task. Rollouts rarely enjoy such optimal conditions and require integration within legacy environments amid competing priorities. That's why proven AI use cases often command a price premium. Iterative business cases and rollouts cannot fully mitigate the risks of unpredictability when buyers and vendors risk buying tokens in advance. So, CIOs need to build extra safety margin into cost-value ratios.

Token supply costs for limited AI capacity are also expected to rise with demand in the near term before they fall in the longer term over the next decade. In the absence of predictable costs, CIOs should prioritize high-value use cases that can absorb variability and still deliver strong outcomes.

Success Measures

Tokenomics metrics typically cover these focus areas:

1. **Supply:** The financial sustainability of AI models and their ecosystems:

- Allocation: Percentage of token consumption or credit burn rate.
- Vesting: How long before investments are turned into value.

2. **Token utility:** The functionality and purpose of token use cases within their ecosystem:

- Staking: Express investments in tokens as the percentage spent on each strategic goal.
- Exchange: Track the conversion of tokens into strategic value with tokens per outcome.

3. **Governance and incentives:** Making it all work in everyone's self-interest:

- Average returns on token spending are a positive incentive but can be inflationary.
- The percentage of nonproductive tokens is a deflationary metric to contain excessive risk taking.

Evidence

¹ [Tokenomics: Quantifying Where Tokens Are Used in Agentic Software Engineering](#), Concordia University.

² [Exclusive: Anthropic acknowledges testing new AI model representing 'step change' in capabilities, after accidental data leak reveals its existence](#), Fortune.

³ [What Are AI Tokens? The Language and Currency Powering Modern AI](#), NVIDIA.

⁴ [The Unifying Language for Technology Value](#), Focus.

Recommended by the Author

Some documents may not be available as part of your current Gartner subscription.

[AI Tokenomics: Measuring LLM Usage in AI-Enabled Applications](#)

[Score Autonomous AI Solutions to Realize Value and Avoid the Token Trap](#)

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